NUMBER THEORY LADDER 6 OF 9

Primes mod 4 and the Gaussian integers

A prime is a sum of two squares exactly when it is 1 mod 4, which is exactly when it splits in the Gaussian integers.

1 / THE RULE AND THE PICTURE

Odd primes fall into two classes, $1 \bmod 4$ and $3 \bmod 4$, and only the first are sums of two squares

1	**2**	**3**	4	**5**	6	**7**	8	9	10
11	12	**13**	14	15	16	**17**	18	**19**	20
21	22	**23**	24	25	26	27	28	**29**	30
31	32	33	34	35	36	**37**	38	39	40
41	42	**43**	44	45	46	**47**	48	49	50
51	52	**53**	54	55	56	57	58	**59**	60
61	62	63	64	65	66	**67**	68	69	70
71	72	**73**	74	75	76	77	78	**79**	80
81	82	**83**	84	85	86	87	88	**89**	90
91	92	93	94	95	96	**97**	98	99	100

1 mod 4 (11) · 3 mod 4 (13) · 2

In the Gaussian integers $\mathbb{Z}[i] = \{a + bi\}$ the norm is $N(a + bi) = a^2 + b^2 = (a + bi)(a - bi)$, so a prime that is a sum of two squares factors:

$$p = a^2 + b^2 = (a + bi)(a - bi)$$

The eleven primes up to 100 that are $1 \bmod 4$ are the ones that factor, the thirteen that are $3 \bmod 4$ stay prime, and $2 = -i(1 + i)^2$ is the single ramified prime.

2 / SPLIT, INERT AND RAMIFIED

Every prime of $\mathbb{Z}$ behaves in one of three ways in $\mathbb{Z}[i]$

type	condition	examples	sum of squares
split	$p \equiv 1 \pmod 4$	$5 = (2 + i)(2 - i)$, $13 = (3 + 2i)(3 - 2i)$	$p = a^2 + b^2$
inert	$p \equiv 3 \pmod 4$	$3, 7, 11, 19$ stay prime	no representation
ramified	$p = 2$	$2 = -i(1 + i)^2$	$2 = 1^2 + 1^2$

The primes $1 \bmod 4$ up to 100 are $5 = 1^2 + 2^2$, $13 = 2^2 + 3^2$, $17 = 1^2 + 4^2$, $29 = 2^2 + 5^2$, $37 = 1^2 + 6^2$, $41 = 4^2 + 5^2$, $53 = 2^2 + 7^2$, $61 = 5^2 + 6^2$, $73 = 3^2 + 8^2$, $89 = 5^2 + 8^2$, $97 = 4^2 + 9^2$.

3 / WHY $1 \bmod 4$ / THE EQUATION $x^2 \equiv -1$

A prime splits exactly when -1 is a square modulo p

p	$x^2 \equiv -1$
5	$2^2 = 4$
13	$5^2 = 25$
17	$4^2 = 16$
29	$12^2 = 144$
7	squares $1, 2, 4$, no
11	squares $1, 3, 4, 5, 9$, no

If $x^2 \equiv -1$ then x has order 4 in the units modulo p, so $4 \mid p - 1$. Conversely for $p \equiv 1 \pmod 4$ the criterion $(-1)^{(p-1)/2} = 1$ makes -1 a square. Then $p \mid x^2 + 1 = (x + i)(x - i)$ while p divides neither factor, so p is not prime in $\mathbb{Z}[i]$, where factorisation is unique, and it splits.

4 / WHICH NUMBERS ARE SUMS OF TWO SQUARES

Primes $3 \bmod 4$ must appear to an even power

A number is a sum of two squares exactly when every prime $\equiv 3 \pmod 4$ in it has an even exponent: $45 = 3^2 \cdot 5 = 6^2 + 3^2$ and $50 = 2 \cdot 5^2 = 7^2 + 1^2 = 5^2 + 5^2$ qualify, while $21 = 3 \cdot 7$ and $63 = 3^2 \cdot 7$ do not. The norm is multiplicative, which gives the identity

$$(a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (ad + bc)^2,$$

so $5 \cdot 13 = 65 = (1 \cdot 2 - 2 \cdot 3)^2 + (1 \cdot 3 + 2 \cdot 2)^2 = 4^2 + 7^2$, and the conjugate choice gives $65 = 8^2 + 1^2$.

5 / THE RACE BETWEEN THE CLASSES

Primes $3 \bmod 4$ lead most of the way, though not for ever

	$1 \bmod 4$	$3 \bmod 4$	
up to 100	11	13	$3 \bmod 4$ ahead
up to 1000	80	87	$3 \bmod 4$ ahead
up to 10 000	609	619	$3 \bmod 4$ ahead

Each class holds half of the primes in the long run, yet the lead belongs to $3 \bmod 4$ almost always, and the first prime at which the $1 \bmod 4$ count overtakes is 26 861.

ASIDE / THE UNITS

The four units $1, i, -1, -i$ are the vertices of the square, so each prime has four associates, $a + bi$ and its rotations.

LINKS TO OTHER SHEETS

The order argument uses *Fermat, Euler and repeating decimals*; the Legendre symbol is in *The Legendre symbol and quadratic reciprocity*; the next sheet is *Sums of two squares and Pythagorean triples*.

Convention. $N(a + bi) = a^2 + b^2$; counts exclude 2; $\mathbb{Z}[i]$ has unique factorisation up to the four units.

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